

Verification Test Procedure

Verification Matrix:

Requirement	Requirements Document	Method	Verification Plan Document	Responsible person	Verification Results Document
1. The DPS shall not suffer damage at temperatures down to -40 °C.	Requirements Specification	Analysis & Demonstration	Verification Test Procedure	Joshua Liao	Verification Results
2. The DPS shall maintain voltage output specifications over a temperature range of 0-70 °C.	Requirements Specification	Analysis & Test	Verification Test Procedure	Joshua Liao	Verification Results
3. The DPS shall have a power efficiency of >60% with 20 mA current load on either of the two outputs.	Requirements Specification	Analysis	Verification Test Procedure	Lyuquan Gao	Verification Results
4. The DPS shall provide 5±0.15 V with current load varying from 0 to 20 mA.	Requirements Specification	Test	Verification Test Procedure	Arno Liu	Verification Results
5. The DPS shall provide 12±0.8 V with current load varying from 0 to 20 mA.	Requirements Specification	Test	Verification Test Procedure	Arno Liu	Verification Results
6. The DPS shall operate with a single voltage input from 0 to 30 V. The input voltage must not be within +/- 10% of either of the output voltages (5 V and 12 V).	Requirements Specification	Analysis	Verification Test Procedure	Chenyi Lu	Verification Results
7. The DPS shall maintain voltage output specifications with an input voltage +/-20% from nominal.	Requirements Specification	Test	Verification Test Procedure	Siyu Liu	Verification Results
8. The DPS shall have flying leads for the connection to the battery.	Requirements Specification	Inspection	Verification Test Procedure	Siyu Liu	Verification Results
9. The DPS shall	Requirements	Inspection	Verification	Chenyi Lu	Verification

have test pins for the output voltages.	Specification		Test Procedure		Results
10. The DPS shall have a mass of no more than 25 g including PCB and components (without leads).	Requirements Specification	Test	Verification Test Procedure	Lyuquan Gao	Verification Results
11. The DPS shall generate maximum 30 mA output current under a short circuit condition on either output.	Requirements Specification	Test	Verification Test Procedure	Arno Liu	Verification Results

Verification procedure:

Requirement 1: The DPS shall not suffer damage at temperatures down to -40°C.

Verification using simulation:

- Test setup:

Set up the DPS circuit as shown in Figure 1 in Multisim, attach ammeters and voltmeters to the 5 V and 12 V outputs respectively.

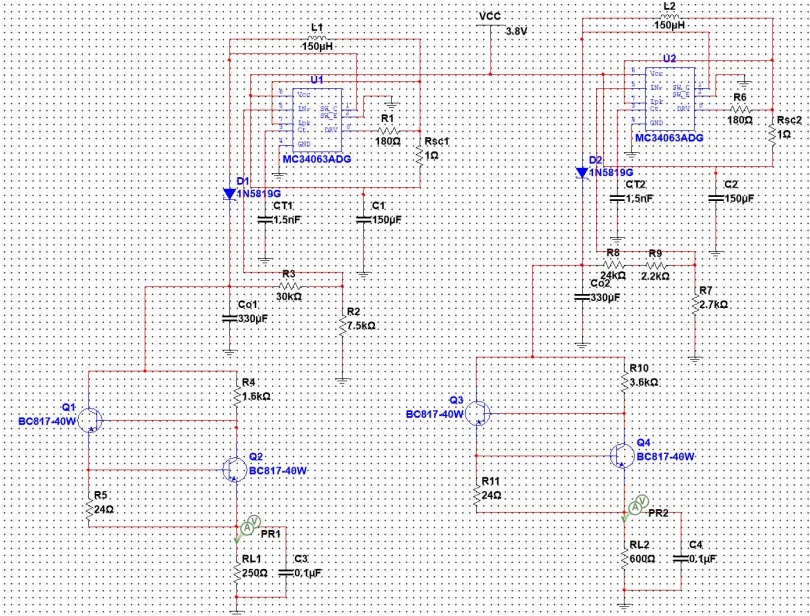


Figure 1: Circuit schematic of the DPS

- Analysis methodology: Analysis
- Procedures:
 - Perform a temperature sweep simulation down to -40 °C on the circuit in Multisim.
 - Check whether the DPS can still maintain the 5 V and 12 V outputs.
- Allowed range of results:

The DPS should output voltages within 5 ± 0.15 V and 12 ± 0.8 V with no physical damage or loss of functionality.

Verification in practice:

- Test setup:
When the DPS is taken out of the thermal chamber for testing, connect the power supply to the input pins of the DPS. Connect a 250 Ω load to the 5 V output pins and a 600 Ω load to the 12 V output pins. Use the oscilloscope to measure the voltage across each load.
- Analysis methodology: Test
- Procedures:
 1. Place the DPS inside a thermal chamber. Gradually decrease the temperature to -40 $^{\circ}\text{C}$.
 2. Use a wide-range scientific thermometer to monitor the temperature inside the chamber. Once -40 $^{\circ}\text{C}$ is reached, carefully take the DPS out using protection gloves or tongs.
 3. Visually inspect the DPS for any signs of physical damage
 4. Connect the DPS to input voltage and connect the output voltages to loads to verify basic functionality.
 5. Leave the DPS for about an hour or until the DPS returns to room temperature. Then repeat step 4.
- Special considerations:
 1. Make sure the DPS is not connected to power when it is placed inside the thermal chamber.
 2. Do not touch the DPS with bare hands.
- Allowed range of results:

The DPS should output voltages within 5 ± 0.15 V and 12 ± 0.8 V with no physical damage or loss of functionality.

Requirement 2: The DPS shall maintain voltage output specifications over a temperature range of 0-70 $^{\circ}\text{C}$.

Verification using simulation:

- Test setup:
Set up the DPS circuit as shown in Figure 2 in Multisim, attach ammeters and voltmeters to the 5 V and 12 V outputs respectively.

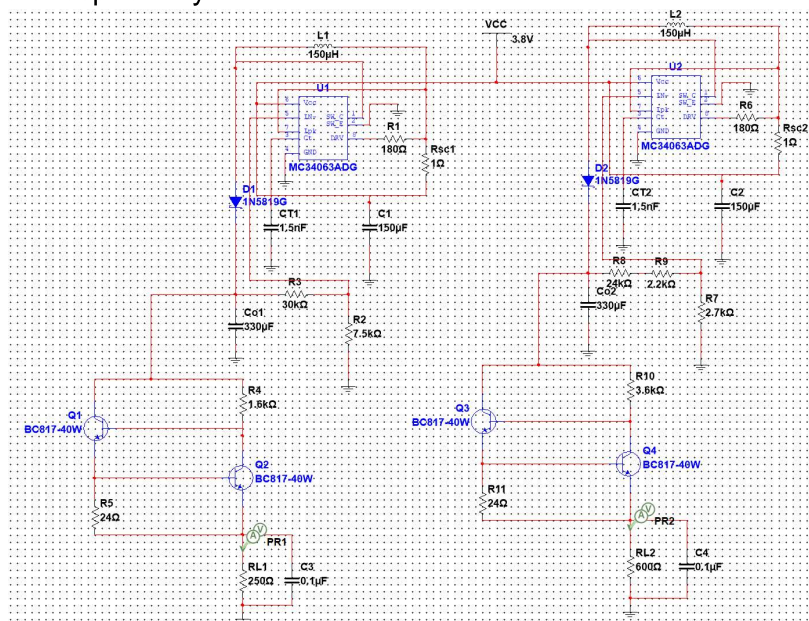


Figure 2: Circuit schematic of the DPS

- Analysis methodology: Analysis

- Procedures:
 1. Perform a temperature sweep simulation from 0 °C to 70 °C on the circuit in Multisim.
 2. Check whether the DPS can still maintain the 5 V and 12 V outputs within the voltage ranges.
- Allowed range of results:

The DPS should output voltages within 5 ± 0.15 V and 12 ± 0.8 V with no physical damage or loss of functionality.

Verification in practice:

- Test setup:

When the DPS is taken out of the thermal chamber for testing, connect the power supply to the input pins of the DPS. Connect a 250 Ω load to the 5 V output pins and a 600 Ω load to the 12 V output pins. Use the oscilloscope to measure the voltage across each load.
- Analysis methodology: Test
- Procedures:
 1. Place the DPS inside a thermal chamber. Gradually increase the temperature from 0 °C to 70 °C in 10 °C increments.
 2. Use a wide-range scientific thermometer to monitor the temperature inside the chamber. At each temperature, carefully take the DPS out using protection gloves or tongs.
 3. Connect the DPS to input voltage and connect the output voltages to loads to verify basic functionality. This process must be quick so that the DPS maintains at the desired temperature.
- Special considerations:
 1. Let the DPS inside the thermal chamber to stabilise at each temperature before taking out for measurements.
 2. Make sure the DPS is not connected to power when it is placed inside the thermal chamber.
 3. Do not touch the DPS with bare hands under test conditions.
- Allowed range of results:

The DPS should output voltages within 5 ± 0.15 V and 12 ± 0.8 V with no physical damage or loss of functionality.

Requirement 3: The DPS shall have a power efficiency of >60% with 20 mA current load on either of the two outputs.

- Test setup:

Use the power supply to provide a 3.8 V to the input pins of the DPS. Connect a 250 Ω load to the 5 V output pins and a 600 Ω load to the 12 V output pins. Connect the oscilloscope in parallel with each load to measure the voltage across each load. Connect the digital multimeter in series with each load to measure the current through each load resistor.
- Analysis methodology: Analysis
- Procedures:
 1. Record the input voltage and input current readings from the power supply.
 2. Record the output voltage across each load resistor using the oscilloscope.
 3. Record the output current through each load resistor using the digital multimeter.
 4. Use the following equation to calculate the overall power efficiency of the DPS:
$$\eta = \frac{(\text{voltage across } 250\Omega \times \text{current through } 250\Omega) + (\text{voltage across } 600\Omega \times \text{current through } 600\Omega)}{\text{input voltage} \times \text{input current}}$$
- Allowed range of results:

Overall power efficiency is greater than 60% and current through each load is approximately 20 mA.

Requirement 4: The DPS shall provide $5 \pm 0.15\text{V}$ with current load varying from 0 to 20mA.

- Test setup:

Use the power supply to provide a 3.8 V to the input pins of the DPS. Connect a load resistor to the 5 V output pins. Connect the oscilloscope in parallel with the load to measure the voltage across the load. Connect the digital multimeter in series with the load to measure the current through the load resistor.

- Analysis methodology: Test

- Procedures:

Adjust the load resistor so that the load current varies from 0 to 20 mA, measure the corresponding output voltage using the oscilloscope.

- Allowed range of results:

5 V output within $\pm 0.15\text{ V}$ for the entire current range.

Requirement 5: The DPS shall provide $12 \pm 0.8\text{V}$ with current load varying from 0 to 20mA.

- Test setup:

Use the power supply to provide a 3.8 V to the input pins of the DPS. Connect a load resistor to the 12 V output pins. Connect the oscilloscope in parallel with the load to measure the voltage across the load. Connect the digital multimeter in series with the load to measure the current through the load resistor.

- Analysis methodology: Test

- Procedures:

Adjust the load resistor so that the load current varies from 0 to 20 mA, measure the corresponding output voltage using the oscilloscope.

- Allowed range of results:

12 V output within $\pm 0.8\text{ V}$ for the entire current range.

Requirement 6: The DPS shall operate with a single voltage input from 0 to 30 V. The input voltage must not be within $\pm 10\%$ of either of the output voltages (5 V and 12 V).

- Test setup:

1. Set up the DPS circuit as shown in Figure 3 in Multisim, attach ammeters and voltmeters to the 5 V and 12 V outputs respectively. Analyse the DPS in Multisim.

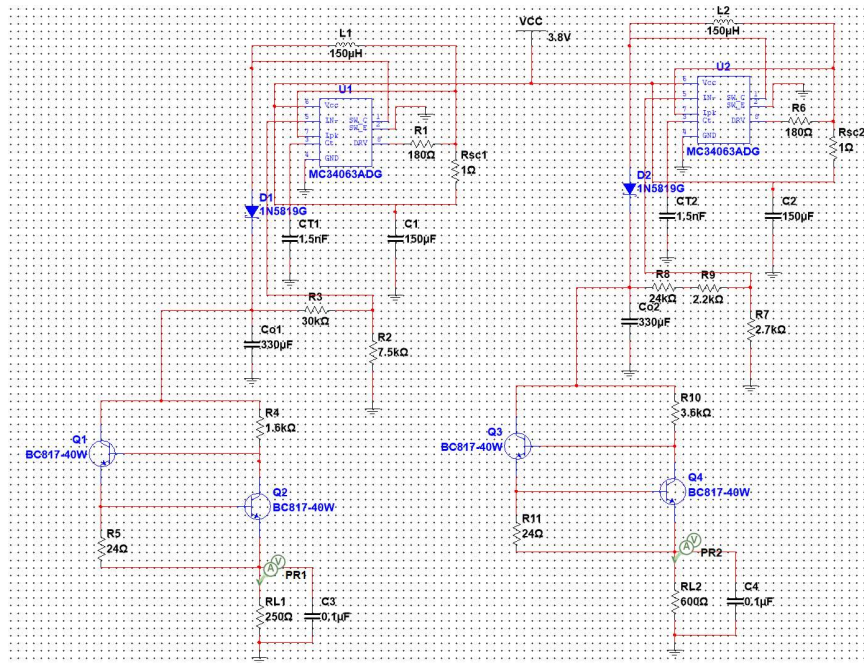


Figure 3: Circuit schematic of the DPS

2. Then, Use the power supply to provide the input voltage based on simulation results to the input pins of the DPS. Connect a 250 Ω load to the 5 V output pins and a 600 Ω load to the 12 V output pins. Connect the oscilloscope in parallel with each load to measure the voltage across each load. Connect the digital multimeter in series with each load to measure the current through each load resistor.

- Analysis methodology: Test

- Procedures:

1. In Multisim, simulate the input voltage range from 0 V to 30 V, avoid input voltage from 4.5 V – 5.5 V and 10.8 V – 13.2 V.
2. Analyse the overall power efficiency, output voltage stability and cost to find the optimal input voltage.
3. Test the optimal input voltage based on simulation results on the actual DPS. Connect the power supply to the input pins of the DPS. Connect a 250 Ω load to the 5 V output pins and a 600 Ω load to the 12 V output pins. Connect the oscilloscope in parallel with each load to measure the voltage across each load. Connect the digital multimeter in series with each load to measure the current through each load resistor.
4. Adjust the input voltage to find the optimal input voltage in practice, make sure the 5 V and 12 V output voltages remain within the specified ranges.

- Special considerations:

The input voltage should be such that it is optimal for efficiency and cost considerations.

- Allowed range of results:

Input voltage within the range 0 V – 4.5 V, 5.5 V – 10.8 V, 13.2 V – 30 V.

Requirement 7: The DPS shall maintain voltage output specifications with an input voltage $\pm 20\%$ from nominal.

- Test setup:

Use the power supply to provide a nominal input voltage of 3.8 V to the input pins of the DPS. Connect a 250 Ω load to the 5 V output pins and a 600 Ω load to the 12 V output pins. Connect the oscilloscope in parallel with each load to measure the voltage across each load. Connect the digital

multimeter in series with each load to measure the current through each load resistor.

- Analysis methodology: Test
- Procedures:

Adjust the 3.8 V input voltage $\pm 20\%$ from nominal in steps, at each input voltage setting, measure the corresponding 5 V and 12 V output voltages across each load resistor using the oscilloscope.

- Allowed range of results:

The 5 V output voltage remains within 4.85 V – 5.15 V; The 12 V output voltage remains within 11.2 V – 12.8 V.

Requirement 8: The DPS shall have flying leads for the connection to the battery.

- Analysis methodology: Inspection
- Procedures:

Visually inspect the DPS to check if the DPS has flying leads for the connection to the battery.

- Allowed range of results:

The DPS has flying leads for the connection to the battery.

Requirement 9: The DPS shall have test pins for the output voltages.

- Analysis methodology: Inspection
- Procedures:

Visually inspect the DPS to check if the DPS has test pins for the output voltages.

- Allowed range of results:

The DPS has test pins for the output voltages.

Requirement 10: The DPS shall have a mass of no more than 25 g including PCB and components (without leads).

- Test setup:

Prepare a laboratory balance as shown in Figure 4 to measure the mass of the DPS.



Figure 4: Picture showing the mass of the DPS

- Analysis methodology: Test

- Procedures:

Weigh the DPS using a laboratory balance.

- Allowed range of results:

The DPS has a mass less than 25 g.

Requirement 11: The DPS shall generate maximum 30 mA output current under a short circuit condition on either output.

- Test setup:

Use the power supply to provide an input voltage of 3.8 V to the input pins of the DPS. Connect the digital multimeter in series with the output pins of each 5 V and 12 V output to measure the short-circuit current through each output.

- Analysis methodology: Test

- Procedures:

Switch on the power supply and record the current reading in each output using the digital multimeter.

- Special considerations:

1. Be prepared to disconnect the power quickly to prevent damage, do not short circuit the outputs over a long time.
2. Monitor the temperature of the DPS.

- Allowed range of results:

Output current in each output does not exceed 30 mA under short circuit conditions.